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DEEPWATER RESONATOR ARRAY FOR SUBSEA NOISE MITIGATION

Abstract

A subsea oil and/or gas facility includes rotating equipment/machinery located subsea and configured for subsea use. The rotating equipment/machinery includes pumping equipment/machinery having one or more pumps, or compression equipment/machinery having one or more compressors, or both, for processing oil and/or gas generated by the subsea oil and/or gas facility. The subsea oil and/or gas facility also includes a noise mitigation system configured for deepwater use and configured to mitigate noise generated by the rotating equipment/machinery. The noise mitigation system includes a deepwater resonator array configured for deepwater use and including an array of resonators filled with gas to mitigate the noise.

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Background/Summary

BACKGROUND

[0001] This section is intended to introduce the reader to various aspects of art that may be related to various aspects of the present disclosure, which are described and/or claimed below. This discussion is believed to be helpful in providing the reader with background information to facilitate a better understanding of the various aspects of the present disclosure. Accordingly, it should be understood that these statements are to be read in this light, and not as admissions of prior art.

[0002] Subsea noise, its sources and impacts are being increasingly scrutinized by non-governmental organizations and regulators. Noise emissions from shipping, defense and other operations and their consequences are being increasingly documented and understood. The development of subsea oil and/or gas facilities and resultant deployment of underwater production trees, machinery and processing equipment on the seabed may generate long-term lower frequency noise that is transmissible over long distances. It may be desirable to reduce the noise output of such equipment.

SUMMARY OF THE INVENTION

[0003] A summary of certain embodiments disclosed herein is set forth below. It should be understood that these aspects are presented merely to provide the reader with a brief summary of these certain embodiments and that these aspects are not intended to limit the scope of this disclosure. Indeed, this disclosure may encompass a variety of aspects that may not be set forth below.

[0004] In accordance with aspects of the disclosure, a subsea oil and/or gas facility includes rotating equipment/machinery located subsea and configured for subsea use. The rotating equipment/machinery includes pumping equipment/machinery having one or more pumps, or compression equipment/machinery having one or more compressors, or both, for processing oil and/or gas generated by the subsea oil and/or gas facility. The subsea oil and/or gas facility also includes a noise mitigation system configured for deepwater use and configured to mitigate noise generated by the rotating equipment/machinery. The noise mitigation system includes a deepwater resonator array configured for deepwater use and including an array of resonators filled with gas to mitigate the noise.

[0005] In accordance with another aspect of the disclosure, a method of mitigating long-term noise produced in a subsea oil and/or gas facility is provided. The subsea oil and/or gas facility includes rotating equipment/machinery located subsea and configured for subsea use. The rotating equipment/machinery includes pumping equipment/machinery having one or more pumps, or compression equipment/machinery having one or more compressors, or both, for processing oil and/or gas generated by the subsea oil and gas facility. The method includes securing a deepwater resonator frame to a support frame of the rotating equipment/machinery, the deepwater resonator frame being attached to a deepwater resonator array including an array of resonator cups formed within resonator panels, the resonator cups being capable of being filled with gas. The method also includes filling the resonator cups with a gas to form resonators, the resonators being configured to mitigate noise generated by the rotating equipment/machinery during operation. Securing the deepwater resonator frame to the support frame places the deepwater resonator array sufficiently proximate the rotating equipment/machinery to enable resonators of the deepwater resonator array to mitigate the noise generated by the rotating equipment/machinery during operation.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0006] The drawings illustrate only example embodiments and are therefore not to be considered limiting in scope, as the example embodiments may admit to other equally effective embodiments. The elements and features shown in the drawings are not necessarily to scale, emphasis instead being placed upon clearly illustrating the principles of the example embodiments. Additionally, certain dimensions or positions may be exaggerated to help visually convey such principles. In the drawings, reference numerals designate like or corresponding, but not necessarily identical, elements.

[0007] FIG. 1 is a block diagram of an example oil and/or gas facility utilizing a deepwater resonator array to mitigate noise associated with subsea equipment/machinery of the facility, in accordance with an embodiment of this disclosure.

[0008] FIG. 2 is a block diagram of an example of the subsea compressor system of FIG. 1 having a deepwater resonator array and a resonator actuation system, in accordance with an embodiment of this disclosure.

[0009] FIG. 3 is an example of the subsea compressor system of FIG. 1 fitted with a deepwater resonator array, in accordance with an embodiment of this disclosure.

[0010] FIG. 4 is a process flow diagram illustrating an example method of mitigating long-term noise produced in a subsea oil and/or gas facility using a deepwater resonator array, in accordance with an embodiment of this disclosure.

DETAILED DESCRIPTION OF THE INVENTION

[0011] As set forth above, it may be desirable to mitigate noise generated by any one or a combination of underwater production trees, machinery and processing equipment on the seabed. In accordance with embodiments of this disclosure, gas bubble based acoustic resonators (e.g., Helmholtz resonators), for example as disclosed in U.S. Pat. Nos. 8,689,935, 9,343,059, 9,488,026, and 9,607,601, which are incorporated by reference herein in their entirety, are adapted from shallow water use (e.g., 50 meters or less) to a deep water application (e.g., at least 1000 m) by encapsulating gas filled resonators arranged in an array-hereinafter referred to as a deepwater resonator array. The array may be, for example, a mesh or network of inverted cups that are at least partially filled with gas, creating gas voids that are specifically tuned to address specific frequencies of concern emitted from the equipment/machinery at a particular installation depth. The equipment/machinery that emits the frequencies of concern may include, by way of non-limiting example, rotating equipment/machinery such as a compressor module or pump module that are a part of a subsea installation.

[0012] The noise mitigation techniques described herein may be associated with prior or concurrent installation of pumping and compression equipment where sound (e.g., low frequency sound, medium frequency sound, high frequency sound, or any combination) is produced. A specific example is the use of the present noise mitigation techniques on a compression system and/or a pumping system deployed in subsea hydrocarbon production facilities (i.e., subsea oil and/or gas production facilities). Further, the noise mitigation techniques described herein may be performed over a long period of time, such that the equipment used to employ such techniques has a design and functional life on the order of, for example, 25-50 years. This is referred to as a “long-term” operational life or deployment as used in this disclosure.

[0013] Thus, in certain embodiments, the present disclosure includes the use of the disclosed deepwater resonator array to mitigate certain types of noise (e.g., low frequency noise, medium frequency noise, high frequency noise, or any combination thereof) produced by subsea oil and/or gas facilities. The deepwater resonator array may be used to mitigate noise associated with, for instance, deep-water pumping and compression equipment. This may be done by installing one or more of the deepwater resonator arrays at specific equipment locations. As discussed in further detail below, a number of approaches may be used to place the resonators in an arrangement that

allows noise mitigation. For example, in some embodiments the resonator arrays may be positioned around (e.g., about, surrounding) at least a portion of the equipment which is the source of the noise to be mitigated (the equipment noise source) via a frame that holds the resonator arrays. Additionally or alternatively, in some embodiments the resonator arrays may be removably coupled to a support structure of the equipment noise source. In a still further embodiment, the resonator arrays may be contained in a flexible jacket or sleeve configured to be placed around the exterior of the equipment noise source.

[0014] The embodiments of this disclosure may encompass a variety of configurations, an example of which is depicted in FIG. 1, which is a block diagram of a subsea oil and/or gas facility **10**. In FIG. 1, the subsea oil and/or gas facility **10** is an oil and/or gas production system (a hydrocarbon production system) utilizing subsea equipment with noise mitigation features, all of which is configured for deepwater use. In the illustrated embodiment, the subsea oil and/or gas facility **10** includes a subsea processing station **12** that processes production fluids (e.g., oil, gas, or a mixture thereof) received from subsea production trees and manifolds **14**. The processed production fluids generated by the subsea processing station **12** are transmitted to topsides equipment **16** by one or more flow lines **18** that cross a water line **20** (e.g., a shoreline or to a topside facility). The topsides equipment **16** may include, by way of non-limiting example, a liquefied natural gas (LNG) plant, a gathering and separation facility, a petroleum-fired power plant, or the like. The subsea oil and/or gas facility **10** may also include certain supporting features that are not shown, such as umbilicals and a local field control station (FCS) or other facility or structure above the waterline. Umbilicals may be used to transmit electrical power and/or signals, hydraulic power and/or signals, or both, between the subsea production trees and manifolds **14**, the subsea processing station **12**, or a combination, and the FCS or other topside facility.

[0015] The subsea processing station **12** as illustrated is positioned on the seabed **22** via one or more structures. Such structures may include one or more mud mats **24** and one or more structural frames **26**. The subsea processing station **12** may include one or more systems (or modules) that process produced fluids, such as by separation, pumping, and/or compression. The one or more systems may include rotating equipment/machinery located subsea and configured for subsea use. Such rotating equipment/machinery may include pumping equipment/machinery (e.g., a pumping module) having one or more pumps, or compression equipment/machinery (e.g., a compression module) having one or more compressors, or both, for processing oil and/or gas generated by the subsea oil and/or gas facility **10**. The illustrated subsea processing station **12** includes a fluid separation system **28** configured to separate produced fluids into its liquid components **30** and gaseous components **32**, a subsea pumping system **34** configured to pump the liquid components **30** and a subsea compression system **36** configured to compress the gaseous components **32**. Pressurized liquid **38** from the subsea pumping system **34** and compressed gas **40** from the subsea compression system **36** are transmitted through the flow lines **18** to the topsides equipment **16**. Although not shown, in some embodiments there may be flowlines that recycle at least a portion of the pressurized liquid **38** and the compressed gas **40** back to the fluid separation system **28**.

[0016] Each of the systems illustrated include their respective components that process the produced fluids in the station **12**. Specifically, the fluid separation system **28** includes separation equipment **42**. The subsea pumping system **34** includes one or more subsea pumps **46** (which may be part of one or more pumping modules); and the subsea compression system **36** includes one or more subsea compressors **50** (which may be part of one or more compression modules). The systems are supported by one or more structural frames **26** and accordingly the one or more mud mats **24** to secure and support the mass of installed equipment.

[0017] To mitigate at least some of the noise generated by the subsea pumps **46** and/or subsea compressors **50**, the station **12** may include a noise mitigation system **54** configured for deepwater use. The noise mitigation system **54** may be positioned in specific locations to disrupt sound waves generated by equipment/machinery noise at its source to avoid sound transmission over long

distances-especially at lower frequencies. Various embodiments of the subsea oil and/or gas facility **10** may utilize any one or a combination of the noise mitigation systems **54** described herein. In the illustrated embodiment, the subsea compression system **36** includes one or more noise mitigation systems **54** located proximate (e.g., around) the subsea compressor(s) **50** (e.g., within the compression module assembly or on the subsea compressor housing).

[0018] During operation, the rotating equipment/machinery such as the pump(s) **46** and compressor(s) **50** produce noise, and it may be desirable to mitigate the noise produced by either or both of these. In accordance with the described embodiments, the subsea compression system **36** and the subsea pumping system **34** may each include one or more noise mitigation systems **54** located proximate (e.g., around) the equipment noise source (e.g., within the module assembly or on the subsea compressor/pump housing).

[0019] In accordance with present embodiments, the noise mitigation system **54** includes a deepwater resonator array **56** configured for deepwater use and comprising an array of resonators (e.g., an array of resonator cups) filled with gas to mitigate noise. The noise mitigation system **54** may also include a gas source **58** to fill the resonator cups. The gas source **58** may include one or more storage tanks, a device or system that performs water electrolysis, or a combination of these. Filling the cups of the resonator array **56** forms resonators that absorb or otherwise disrupt noise generated during operation of the subsea pump(s) **46** and/or the subsea compressor(s) **50**. Although the components of the noise mitigation system **54** are only illustrated for the subsea compression system **36**, it should be noted that in various different embodiments these components may be replicated for the subsea pump system **34**, or may be utilized for both the subsea compression system **36** and the subsea pumping system **34**, or may only be associated with the subsea pump system **34**. In one particular embodiment, the subsea oil and/or gas facility **10** only includes the noise mitigation system **54** associated with the subsea compression system **36**. The noise mitigation system **54** may be permanently installed for the operating life (which may be on the order of 25-50 years) of the rotating equipment/machinery for which it is mitigating noise.

[0020] The deepwater resonator array **56** of this disclosure may have any appropriate shape and material construction that is suitable for mitigating noise generated by the subsea equipment/machinery mentioned above and may depend on the configuration utilized. As the deepwater resonator array **56** is configured to be used in a subsea installation, the deepwater resonator array **56** may have a designed noise mitigation level at a water depth of at least 100 meters, at least 500 meters, at least 1000 meters, at least 1250 meters, at least 1500 meters, at least 2000 meters, or at least 2500 meters; for example between 1000 meters and 1500 meters, or between 2000 meters and 3200 meters. The deepwater resonator array **56** may be configured to mitigate noise at an operational (“dynamic”) design temperature of 2° C. to 30° C. Such a temperature range may allow the deepwater resonator array **56** to be pre-installed on certain types of equipment before final deployment at their ultimate subsea location. In one non-limiting embodiment, the deepwater resonator array **56** may be configured to mitigate noise at an operational temperature of between 25° C. and 30° C. at an operational depth of between 100 m and 150 m. In another non-limiting embodiment, the deepwater resonator array **56** may be configured to mitigate noise at an operational temperature of between 3° C. and 5° C. at an operational depth of between 1200 m and 1500 m. In a further non-limiting embodiment, the deepwater resonator array **56** may be configured to mitigate noise at an operational temperature of between 3° C. and 5° C. at an operational depth of up to 3200 m.

[0021] The configuration of the deepwater resonator array **56** may include shape, thickness, and material construction selected such that the deepwater resonator array is configured to attenuate noise having particular frequencies, such as the low frequency portion of the spectrum, the medium frequency portion of the spectrum, the high frequency portion of the spectrum, or any combination thereof. Generally, the deepwater resonator array **56** may be configured to attenuate noise having a frequency ranging from 5 Hz to 10,000 Hz. In a first embodiment, at least one deepwater resonator

array **56** is configured to mitigate noise output from one or more pumps, the noise having a range of between 40 Hz and 700 Hz. In a second embodiment, at least one deepwater resonator array **56** is configured to mitigate noise output from one or more compressors in a range of between 600 Hz and 2500 Hz.

[0022] The components of or associated with the noise mitigation system **54** may be further appreciated with respect to FIGS. 2-4. In FIG. 2, the subsea oil and/or gas facility **10** as illustrated includes a subsea compression system **36** having the noise mitigation system **54**. The subsea compression system **36** may include a compression module **66** that the noise mitigation system **54** surrounds and/or to which the noise mitigation system **54** is attached. The compression module **66** may include a compression module support structure **72**, one or more compressors **50**, and other equipment **68** (e.g., piping, valves, instruments, cathodic protection, etc.). The module support structure **72** may be constructed from a material that is selected for use in a deepwater setting and may structurally support the compressor(s) **50** and other equipment **68** of the compression module **66**.

[0023] In the illustrated embodiment, the noise mitigation system **54** has the resonator array **56** configured for deepwater use and having an array of resonators filled with gas to mitigate noise generated by the subsea compression system **36** (e.g., by the compression module **66**). The resonator array **56** is generally supported by a resonator support structure **70**, which may include a frame constructed from a material that is selected for use in a deepwater setting. The resonator support structure **70**, in some embodiments, may surround the compression module **66**, such as by surrounding the support structure **72** (e.g., support frame) of the compression module **66**. The resonator support structure **70** (e.g., frame) of the noise mitigation system **54** may be connected to the support frame **72** of the compression module **66**. Such a configuration places the resonator array **56** in position to mitigate the noise generated by the subsea compression module **66**. As discussed in further detail with respect to FIG. 3, the frame **70** may include one more sections that place portions of the resonator array **56** in particular locations, orientations, and so forth.

[0024] As shown in FIG. 2, the noise mitigation system **54** may also include a resonator actuation system **74**, which may include an actuation mechanism constructed from materials that are selected for use in a deepwater setting. The resonator actuation system **74** is coupled to the resonator support structure **70** and may include one or more devices (e.g., actuators) for selectively moving one or more parts (e.g., panels or groups of panels) of the resonator array **56** with respect to the compressor(s) **50** of the subsea compression system **36**. The resonator actuation system **74** may be a mechanical actuation system including, for example, a winch, an interface comprising a receptacle for a torque tool, and the like. In another example, the resonator actuation system **74** may be an electrical actuation system including, for example, an electric motor powered via a cable extending from topsides equipment. In another example, the resonator actuation system **74** may be a hydraulic actuation system including, for example, a hydraulic piston. Any combination of the above types of actuation mechanisms may be used in the resonator actuation system **74** of FIG. 2.

[0025] As shown in FIG. 2, the noise mitigation system **54** may also include a resonator monitoring and control system **76**. As discussed above, the noise mitigation system **54** may include a gas source **58** used for filling resonator cups of the resonator array **56**. The resonator control system **76** may be communicatively coupled with the gas source **58** and configured to control the gas source **58** in response to sensor signals and/or commands received from topsides equipment of the subsea oil and/or gas facility **10**. In one example, the resonator control system **76** may be located subsea proximate the noise mitigation system **54**. In another example, the resonator control system **76** may be located within or proximate the topsides equipment. The resonator control system **76** may be communicatively coupled to the gas source **58**, for example, via an electric, hydraulic, or fiber optic connection, to communicate control signals to initiate a cup refill sequence to output gas from the gas source **58** into the resonator array **56**.

[0026] The resonator control system **76** generally includes at least one processing component **78**

and at least one memory component **80**. The at least one memory component **80** is a non-transitory computer-readable medium containing instructions that, when executed by the at least one processing component, cause the resonator control system **76** to perform one or more operations. For example, the instructions in the at least one memory component **80**, when executed by the at least one processing component, may cause the resonator control system **76** to output control signal(s) to the resonator actuation system **74** to move one or more portions of the resonator array **56** in response to commands from the topsides equipment received by the at least one processing component **78**. The at least one memory component **80** may include instructions that, when executed by the at least one processing component **78**, cause the resonator control system **76** to output control signal(s) to the gas source **58** to output gas for filling resonator cups of the resonator array **56** in response to sensor signals and/or commands from the topsides equipment received by the at least one processing component **78**. The “sensor signals” may include either direct measurements (e.g., measuring the gas fill level in the resonator cups) and/or indirect measurements (e.g., overall station noise monitoring) indicative of the current gas fill level in the resonator array **56**.

[0027] In some embodiments, the resonator monitoring and control system **76** may be communicatively coupled with the resonator actuation system **74** and configured to control the resonator actuation system **74** in response to commands received from topsides equipment of the subsea oil and/or gas facility **10**. The resonator control system **76** is communicatively coupled to the resonator actuation system **74**, for example, via an electric, hydraulic, or fiber optic connection, to communicate control signals for operating the resonator actuation system **74**. The at least one memory component **80** may include instructions that, when executed by the at least one processing component **78**, cause the resonator control system **76** to output control signal(s) to the resonator actuation system **74** to move one or more portions of the resonator array **56** in response to commands from the topsides equipment received by the at least one processing component **78**. Such movement of the resonator array **56** by the resonator actuation system **74** may facilitate deployment (e.g., unfolding/independently lowering) of resonators to provide complete coverage of the compression module **66**. Additionally or alternatively, the movement of the resonator array **56** by the resonator actuation system **74** may facilitate access for periodic intervention/maintenance activities on the compression module **66** without having to remove the entire noise mitigation system **54**, for example, by opening roof hatches to allow a subsea control module to be removed from the top of the compression module **66**. Although FIG. **2** shows the noise mitigation system **54** being used in the context of a subsea compression system **36**, it should be understood that a similar arrangement of the noise mitigation system **54** (e.g., including resonator support structure **70**, resonator actuation system **74**, gas source **58**, and/or resonator control system **76**) may be used with a subsea pumping system (e.g., **34** of FIG. **1**) or any other rotating equipment/machinery of a subsea oil and/or gas facility.

[0028] FIG. **3** shows an example subsea compression system **36** having a noise mitigation system **54** disposed around the rotating equipment/machinery (i.e., compression module **66** having one or more compressors **50**). As illustrated, the noise mitigation system **54** includes a deepwater resonator array **56** and resonator array support structure (e.g., frame) **70**. The frame **70** of the noise mitigation system **54** surrounds the compression module **66**. The compression module **66** (and likewise a pump module having one or more pumps **46** as described above with reference to FIG. **1**) may have one or more retrievable elements. As described above, the compression module **66** may include at least one compressor **50**, as well as other equipment such as dedicated control module(s), piping, valves, instruments, cathodic protection, etc. Similarly, a pump module may include at least one pump (**46**), as well as other equipment such as valves and piping, etc. The compression module **66** is the main element within the compression system **36**, and similarly the pump module is the main element within the pump system **34**.

[0029] As shown in the illustrated embodiment, the deepwater resonator array **56** may include a

plurality of panels **110** in which resonator cups **112** are formed. The plurality of panels **110** are attached to the frame **70** such that the deepwater resonator array **56** surrounds the compression module **66** on at least one side. In the illustrated embodiment, the panels **110** are attached to the frame **70** such that the deepwater resonator array **56** surrounds the compression module **66** on all four sides and the top (but not the bottom, which is located on a structure at the seabed). The frame **70** is configured to hold the plurality of panels **110** in a noise mitigation orientation in which the resonator cups **112** on the panels **110** each have an open end oriented toward the seabed to enable capture of gas for noise attenuation.

[0030] The deepwater resonator array **56** may include multiple groups of panels **110**. The panels **110** in each group of panels **110** are connected to each other, either rigidly through the frame **70** or movably via a resonator actuation system coupled to the frame **70** to allow periodic access and/or intervention to the retrievable elements within the compression module **66**.

[0031] The frame **70** of the noise mitigation system **54** may be positioned around the compression module support structure (e.g., module support frame) **72** of the compression module **66**. The frame **70** of the noise mitigation system **54** may be connected to the module support frame **72** of the compression module **66**. In some embodiments, the frame **70** may include multiple sections that are configured to hold different panels **110** (or different groups of panels **110**). For example, the frame **70** of the noise mitigation system **54** may include a first section **114** and a second section **116**, the first section **114** configured to hold at least a first panel **110A** of the plurality of panels **110** and the second section **116** configured to hold at least a second panel **110B** of the plurality of panels **110**. The first section **114** may be openable in certain embodiments, as described further below. In addition, as shown, the frame **70** of the noise mitigation system **54** may include a third section **118** configured to hold at least a third panel **110C** of the plurality of panels **110**.

[0032] In some embodiments, the different sections **114/116/118** may be configured to hold their respective panels **110A/110B/110C** at different relative orientations with respect to the compression module **66** and/or the seabed. For example, the first section **114** may be configured to hold the first panel(s) **110A** in a first orientation, while the second section **116** may be configured to hold the second panel(s) **110B** in a second orientation different from the first orientation. This may facilitate periodic access for intervention and/or maintenance of the compression module **66** as required while avoiding removal of the entire resonator array **70**.

[0033] As discussed above, the noise mitigation system **54** may include at least one resonator actuation system (e.g., **74** of FIG. 2). The resonator actuation system(s) may be incorporated into the frame **70** and/or connected to one or more sections **114/116/118** of the frame **70**. Each resonator actuation system may be configured to move the first panel **110A**, the second panel **110B**, the third panel **110C**, or a combination thereof. In the illustrated embodiment, the noise mitigation system **54** includes multiple resonator actuation systems (e.g., each for moving a different group of panels).

[0034] In some embodiments, the resonator actuation system may be coupled to the frame **70** and configured to selectively move one or more sections of the frame **70** (and the attached panels **110**) relative to the compression module **66**. For example, in FIG. 3, one resonator actuation system may be configured to selectively move the first section **114** (and its panels **110A**) away from the compression module **66**. In some embodiments, the resonator actuation system may be coupled to the frame **70** and configured to selectively move one or more panels (or groups of panels) **110** relative to the frame **70** and the compression module **66**. For example, in FIG. 3, resonator actuation systems may be configured to selectively move the second panels **110B** relative to the stationary second section **116** of the frame **70** and the third panels **110C** relative to the stationary third section **118** of the frame **70**.

[0035] The resonator actuation system(s) may provide one or more movable sets of panels **110** within the subsea compression system **36**. The resonator actuation system(s) may facilitate installation or periodic intervention on elements within the compression module **66** (e.g., a set of panels **110B** that could be moved via cables) with primary actuation from a remotely operated

vehicle (ROV) external to the compression system **36**. To that end, the resonator actuation system may include an ROV interface **122** configured to allow an ROV to cause any one or a combination of the first panel **110A**, the second panel **110B**, and the third panel **110C**, to move. The ROV interface **122** may include a receptacle for a torque tool of an ROV. In some embodiments, the resonator actuation system may include one or more winches **120** attached (e.g., via cables) to any one or a combination of the first panel **110A**, the second panel **110B**, and the third panel **110C**. The resonator actuation systems shown in FIG. **3** are mechanical actuation systems. However, as discussed above, other types of actuation systems may be used in other embodiments without departing from the scope of this disclosure.

[0036] In some embodiments, one or more sections of the frame **70** of the noise mitigation system **54** may be selectively movable with respect to the compression module **66**. As shown in FIG. **3**, for example, the first section **114** of the frame **70** may be hinged or retractable to allow the first section **114** to be rotated away from the compression module **66** and thereby allow access to intervene on retrievable elements within the compression module **66** while the remaining sections (e.g., **116**, **118**, etc.) of the frame **70** remain stationary. A resonator actuation system may cause this movement of the first section **114** of the frame **70**. The first panel **110A** may be fixedly attached to the first section **114** of the frame **70** such that the first panel **110A** is unable to move relative to the first section **114** of the frame **70**. As such, during movement of the first section **114** of the frame **70**, the panel (or group of panels) **110A** attached to the first section **114** remains in a substantially stationary position and orientation with respect to one another. As such, movement of the section **114** of the frame **70** from a noise mitigation position to a maintenance position moves all panels **110A** away from the compression module **66** simultaneously.

[0037] One or more other sections of the frame **70** may remain stationary with respect to the compression module **66** while allowing for their attached panels **110** to be retracted. For example, in FIG. **3**, the second panel(s) **110B** of the deepwater resonator array **56** are attached to each other and to the second section **116** of the frame **70** via a resonator actuation system. The resonator actuation system includes a group of cables that can be pulled upward on winches **120** attached to the second section **116** of the frame **70** to retract the second panel(s) **110B**. The retractable cables may be attached to both opposing ends of each second panel **110B**. As illustrated, the group of panels **110B** attached to the second section **116** of the frame **70** may be pulled upward and collapsed like a set of blinds, thereby allowing access to the compression module **66** while one or more other sections of the frame **70** remain stationary. As such, the second panel **110B** may be attached to the second section **116** of the frame **70** in a configuration allowing movement in a direction that retracts the second panel **110B** away from the compression module **66**.

[0038] As another example, the third panel **110C** (or group of panels) of the deepwater resonator array **56** are attached to each other and to the third section **118** of the frame **70** via a resonator actuation system. The resonator actuation system includes one or more cables that can be pulled upward on a winch **120** attached to the third section **118** of the frame **70** to retract the third panel(s) **110C**. The retractable cable(s) may be attached to only one side of each third panel **110C**, as shown. As illustrated, the group of panels **110C** attached to the third section **118** of the frame **70** may be rotated and collapsed on themselves as their ends are pulled upward to facilitate deployment of the panels **110C** onto the compression module **66** or to allow access to retrievable elements within the compression module **66** while one or more other sections of the frame **70** remain stationary. As such, the third panel **110C** may be attached to the third section **118** of the frame **70** in a configuration allowing movement in a direction that retracts the third panel **110C** away from the compression module **66**.

[0039] Although FIG. **3** shows the noise mitigation system **54** being used in the context of a subsea compression system **36**, it should be understood that a similar arrangement of the noise mitigation system **54** (e.g., including multiple panels **110**, frame **70** with or without different sections, resonator actuation system(s), etc.) may be used with a subsea pumping system (e.g., **34** of FIG. **1**)

or any other rotating equipment/machinery of a subsea oil and/or gas facility.

[0040] FIG. 4 is a process flow diagram illustrating an example method **400** of mitigating long-term noise produced in a subsea oil and/or gas facility using a deepwater resonator array. As discussed above, the subsea oil and/or gas facility includes rotating equipment/machinery (e.g., pumping equipment/machinery having one or more pumps, compression equipment/machinery having one or more compressors, or both) for processing oil and/or gas generated by the subsea oil and/or gas facility. The method **400** includes blocks **402-408** shown in FIG. 4. It should be noted that in certain embodiments, the method **400** may not include every step shown in FIG. 4, the method **400** may include additional steps not shown in FIG. 4, and/or the steps of the method **400** may be performed in different orders than shown in FIG. 4, without departing from the scope of this disclosure.

[0041] At block **402**, the method **400** includes securing a deepwater resonator frame to a support frame of the rotating equipment/machinery. The deepwater resonator frame is attached to a deepwater resonator array including an array of resonator cups formed within resonator panels. The resonator cups are capable of being filled with gas. Securing (**402**) the deepwater resonator frame to the support frame places the deepwater resonator array sufficiently proximate the rotating equipment/machinery to enable resonators of the deepwater resonator array to mitigate the noise generated by the rotating equipment/machinery during operation. In some embodiments, securing (**402**) the deepwater resonator frame to the support frame is done after deployment of the rotating equipment/machinery in a deepwater environment. In other embodiments, securing (**402**) the deepwater resonator frame to the support frame is done before deployment of the rotating equipment/machinery in a deepwater environment. At block **404**, the method **400** includes filling the resonator cups with a gas to form resonators. The resonators are configured to mitigate noise generated by the rotating equipment/machinery during operation.

[0042] At block **406**, the method **400** may include moving at least one resonator panel of the resonator panels from a noise mitigation position to a maintenance position. This movement (**406**) may be performed via an actuation mechanism integrated into the deepwater resonator frame that allows the resonator panels to move relative to the deepwater resonator frame. The maintenance position allows greater access to the rotating equipment/machinery from outside of the deepwater resonator frame than the noise mitigation position. At block **408**, the method **400** may include returning the at least one resonator panel from the maintenance position to the noise mitigation position without moving the deepwater resonator frame. The resonator panel(s) may be returned (**408**) from the maintenance position to the noise mitigation position after performing maintenance, or after an inspection on the rotating equipment/machinery.

[0043] It will be understood that various modifications may be made to the embodiments disclosed herein. Therefore, the above description should not be construed as limiting, but merely as exemplifications of example embodiments. For example, the functions described above and implemented as the best mode for operating the present invention are for illustration purposes only. Other arrangements and methods may be implemented by those skilled in the art without departing from the scope and spirit of this invention. Moreover, those skilled in the art will envision other modifications within the scope and spirit of the claims appended hereto.

Claims

1. A subsea oil and/or gas facility comprising: rotating equipment/machinery located subsea and configured for subsea use, the rotating equipment/machinery comprising pumping equipment/machinery having one or more pumps, or compression equipment/machinery having one or more compressors, or both, for processing oil and/or gas generated by the subsea oil and/or gas facility; and a noise mitigation system configured for deepwater use and configured to mitigate noise generated by the rotating equipment/machinery, wherein the noise mitigation system

comprises a deepwater resonator array configured for deepwater use and comprising an array of resonators filled with gas to mitigate the noise.

2. The subsea oil and/or gas facility of claim 1, wherein the noise mitigation system further comprises a frame surrounding the rotating equipment/machinery, and wherein the deepwater resonator array comprises a plurality of panels in which resonator cups are formed, the plurality of panels being attached to the frame such that the deepwater resonator array surrounds the rotating equipment/machinery on at least one side, and wherein the frame is configured to hold the plurality of panels in a noise mitigation orientation in which the resonator cups have an open end oriented toward the seabed to enable capture of gas for noise attenuation.

3. The subsea oil and/or gas facility of claim 2, wherein the frame of the noise mitigation system is positioned around a module support frame of the rotating equipment/machinery.

4. The subsea oil and/or gas facility of claim 3, wherein the frame of the noise mitigation system is connected to the module support frame of the rotating equipment/machinery.

5. The subsea oil and/or gas facility of claim 2, wherein the frame of the noise mitigation system comprises a first section and a second section, the first section being configured to hold a first panel of the plurality of panels and the second section being configured to hold a second panel of the plurality of panels.

6. The subsea oil and/or gas facility of claim 5, wherein the first section is configured to hold the first panel of the plurality of panels in a first orientation and the second section is configured to hold the second panel of the plurality of panels in a second orientation different from the first.

7. The subsea oil and/or gas facility of claim 5, wherein the first section is hinged or retractable to allow the first section to be rotated away from the rotating equipment/machinery and thereby allow access to the rotating equipment/machinery while remaining sections of the frame remain stationary.

8. The subsea oil and/or gas facility of claim 5, wherein the first panel is fixedly attached to the first section such that the first panel is unable to move relative to the first section of the frame, and the second panel is attached to the second section in a configuration allowing movement in a direction that retracts the second panel away from the rotating equipment/machinery.

9. The subsea oil and/or gas facility of claim 5, wherein the noise mitigation system comprises a resonator actuation system configured to move either or both of the first panel and the second panel.

10. The subsea oil and/or gas facility of claim 9, wherein the resonator actuation system comprises one or more winches attached to either or both of the first panel and the second panel.

11. The subsea oil and/or gas facility of claim 9, wherein the resonator actuation system comprises a remotely operated vehicle (ROV) interface configured to allow an ROV to cause the first panel, the second panel, or both, to move.

12. The subsea oil and/or gas facility of claim 11, wherein the ROV interface comprises a receptacle for a torque tool of the ROV.

13. The subsea oil and/or gas facility of claim 9, wherein the resonator actuation system is coupled to the frame and configured to selectively move one or both of the first section and the second section of the frame relative to the rotating equipment/machinery.

14. The subsea oil and/or gas facility of claim 1, wherein the noise mitigation system further comprises: a gas source; and a control system communicatively coupled with the gas source and configured to control the gas source in response to sensor signals and/or commands received from topsides equipment of the subsea oil and/or gas facility.

15. The subsea oil and/or gas facility of claim 1, wherein the noise mitigation system is permanently installed for the operating life of the rotating equipment/machinery.

16. A method of mitigating long-term noise produced in a subsea oil and/or gas facility comprising rotating equipment/machinery located subsea and configured for subsea use, the rotating equipment/machinery comprising pumping equipment/machinery having one or more pumps, or

compression equipment/machinery having one or more compressors, or both, for processing oil and/or gas generated by the subsea oil and gas facility, the method comprising: securing a deepwater resonator frame to a module support frame of the rotating equipment/machinery, the deepwater resonator frame being attached to a deepwater resonator array comprising an array of resonator cups formed within resonator panels, the resonator cups being capable of being filled with gas; and filling the resonator cups with a gas to form resonators, the resonators being configured to mitigate noise generated by the rotating equipment/machinery during operation; and wherein securing the deepwater resonator frame to the module support frame places the deepwater resonator array sufficiently proximate the rotating equipment/machinery to enable resonators of the deepwater resonator array to mitigate the noise generated by the rotating equipment/machinery during operation.

17. The method of claim 16, comprising: moving at least one resonator panel of the resonator panels from an initial position to a noise mitigation position via an actuation mechanism integrated into the deepwater resonator frame that allows the resonator panels to move relative to the deepwater resonator frame, the initial position maintaining the at least one resonator panel in a collapsed position proximate one or more other resonator panels while the deepwater resonator frame is secured to the module support frame.

18. The method of claim 16, comprising: moving at least one resonator panel of the resonator panels from a noise mitigation position to a maintenance position via an actuation mechanism integrated into the deepwater resonator frame that allows the resonator panels to move relative to the deepwater resonator frame, the maintenance position allowing greater access to the rotating equipment/machinery from outside of the deepwater resonator frame than the noise mitigation position; and returning the at least one resonator panel from the maintenance position to the noise mitigation position without moving the deepwater resonator frame.

19. The method of claim 18, comprising returning the at least one resonator panel from the maintenance position to the noise mitigation position after performing maintenance, or an inspection on the rotating equipment/machinery.

20. The method of claim 16, wherein securing the deepwater resonator frame to the module support frame is done after deployment of the rotating equipment/machinery in a deepwater environment.

21. The method of claim 16, wherein securing the deepwater resonator frame to the module support frame is done before deployment of the rotating equipment/machinery in a deepwater environment.
